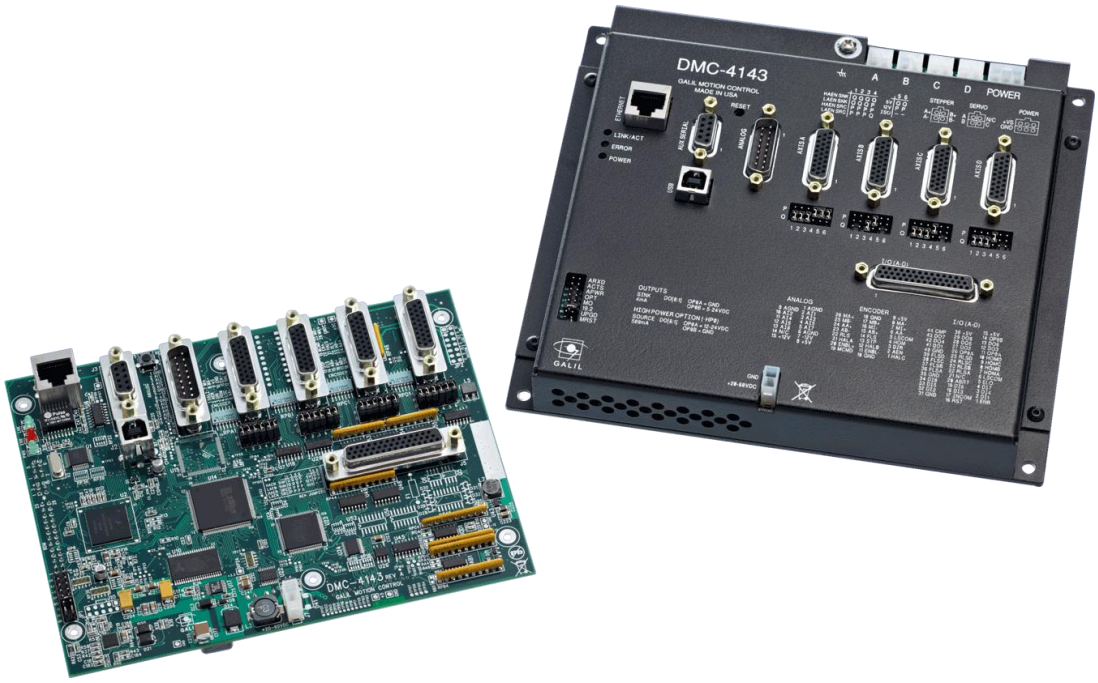
Galil Motion Control



DMC – 41x3

Datasheet

Galil Motion Control 270 Technology Way, Rocklin, CA 1-800-377-6329

Product Description

The DMC-41x3 motion controller is one of Galil's latest generation, standalone motion controllers. Compared with the previous generation's DMC-21x3 controller, the DMC-41x3 offers the following enhancements: 100BASE-T Ethernet, auxiliary RS232 port, USB port, uncommitted analog inputs, up to 15 MHz encoder frequencies, increased program memory, sampling higher frequencies. The DMC-41x3 also accommodates the same stepper and servo motor drives used in the DMC-40x0 series and allows two 4-axis 750 W drives to be installed in the 8 axis controller package.

The DMC-41x3 is available as a box-level or card-level motion controller. The unit operates stand-alone or interfaces to a PC with Ethernet 10/100BASE-T, or USB. It includes optically isolated I/O in addition to analog inputs and outputs. The DMC-41x3 controller and drive unit accepts power from a single 20-80 VDC source. The DMC-41x3 is available in one through eight axis formats, and each axis is user-configurable for stepper or servo motor operation. Standard programming features include PID compensation with velocity and acceleration feedforward, multitasking for simultaneously running up to eight programs, and I/O processing for synchronizing motion with external events.

Features

- Ethernet 10/100Base-T port; (1) USB - main; (1) RS-232 - aux.
- Optional plug-in amplifier boards for driving stepper, brush, or brushless servo motors up to 750 Watts. Or, connect to external drives of any power range.
- User configurable for stepper or servo motors on any combination of axes. Optional firmware for piezo-ceramic motors. Configurable for sinusoidal commutation.
- Accepts up to 15MHz encoder frequencies for servos and outputs up to 3MHz for steppers.
- PID compensation with velocity and acceleration feedforward, integration limits, notch filter and low-pass filter.
- Modes of motion include jogging, point-to-point positioning, contouring, PVT, linear and circular interpolation, electronic gearing and ECAM. Ellipse scaling, slow-down around corners, infinite segment feed and feedrate override.
- Over 200 English-like commands including conditional statements and event triggers.
- Non-volatile memory for programs, variables and arrays. Multitasking for concurrent execution of up to eight programs.
- Optically isolated home input and forward and reverse end-of-travel limits for every axis.
- Uncommitted, isolated inputs and isolated outputs:
  - 1- 4-axis models: 8 inputs and 8 outputs (4mA default; 25mA or 500mA option)
  - 5- 8-axis models: 16 inputs and 16 outputs (4mA default; 25mA or 500mA option)
- High speed position latch for each axis and output compare.
- 8 uncommitted analog inputs.
- Dual encoder inputs for each servo axis.
- Accepts single 20-80 VDC input.
- Available as card-level or with metal enclosure.
- Communication drivers for Windows, Mac OSX, and Linux.
- Custom hardware and firmware options available.

Motion Controller	
Processor	RISC-based clock multiplying processor with DSP functions
Communication	10/100BASE-T Ethernet with Auto MDIX, USB port – main, RS232 port - aux
Program memory size	4000 lines x 80 characters
# of Variables	510
# of Arrays	24000 array elements in 30 arrays
Position Range	32-bit, automatic rollover
Maximum Velocity	15 million counts/s
Maximum Acceleration	1 billion counts/s^2

Power and Mechanical	
Power requirements	20-80 V_{DC}
Operational temperature	0 – 70º C
Humidity	20 – 95 % RH, non-condensing
Dimensions	**1-4 axes models:** 8.05” x 7.25” x 1.5” **5-8 axes models:** 11.5” x 7.25” x 1.5”

3137106

RoHS Compliant

Configurable Filter Features
Proportional
Torque limit
Backlash compensation
Integral
Offset
Profile filtering
Derivative
Feed-forward acceleration
Low-pass filter (Pole)
Notch
Dual-loop feedback mode
Feed-forward velocity

Modes of Motion	
Position Relative & Position Absolute	Absolute and relative positioning following a trapezoidal velocity profile. Phase correction and profile smoothing available.
Jogging	Velocity control where no final endpoint is prescribed.
Vector Mode	2D motion path consisting of linear and arc segments. Motion along the path is continuous at the prescribed vector speed even at transitions between linear and circular segments.
Linear Interpolation	1-8 axes of coordinated linear profiling.
Gearing & Gantry Mode	Electronic gearing and gantry mode with ramped gearing.
Electronic camming (ECAM)	Following an arbitrary trajectory based upon a master encoder position.
Contour	Allows any arbitrary profile and any set of axes to be prescribed.
PVT	Motion path described in incremental position, velocity, and change of time.

Minimum Servo Update Rate	
# of axes	**Standard Firmware**
1-2	125 usec, 8 kHz
3-4	250 usec, 4 kHz
5-6	375 usec, 2.6 kHz
7-8	500 usec, 2 kHz

General Purpose I/O				
	Number of I/O		Voltage	Details
	1-4 axis	5-8 axis		
Opto-isolated inputs[1]	8	16	5-24 V_{DC}	Can be configured for use as high-speed latch (position capture).
Opto-isolated outputs	8	16	5-24 V_{DC}	500mA Sourcing, can be configured as brake output.
Analog Inputs	8	8	±10, ±5, 0-5, 0-10 V	12-bit, 16-bit optional, can be used as position feedback

Feature Specific I/O				
	Number of I/O		Description	Details
	1-4 axis	5-8 axis		
Reverse/Forward Limit Switches	per Axis		5-24 V_{DC} Opto-isolated	
Home Input	per Axis		5-24 V_{DC}, Opto-isolated	
Amplifier Enable Output	per Axis		+5, +12V_{DC} controller powered or 5-24V_{DC} Opto-isolated	Configurable with jumpers.
Stepper (Step/Dir signals)	per Axis		0-5 V_{DC} Step/Dir TTL Signal	3 MHz max output
Servo control (Motor command line)	per Axis		±10V analog output	16-bit resolution
Quadrature Encoder Inputs	2 per Axis[1]		+/-12V_{DC} or TTL	15 MHz input max
Hall inputs	per Axis		3x 0-5V TTL inputs	When equipped with some AMP Modules
Abort	1		5-24V_{DC} Opto-isolated	
Reset	1		5-24V_{DC} Opto-isolated	
Electronic lock-out	1		5-24V_{DC} Opto-isolated	When equipped with AMP Modules
Output compare	1	2	0-5V TTL	Also known as pulse on position
Error out	1		0-5V TTL	

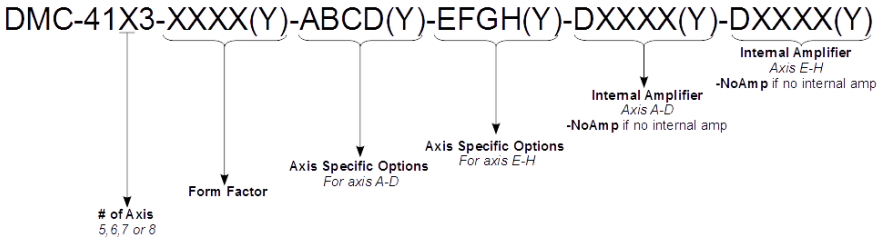
[1] Each unused auxiliary encoder can be used as 2 additional digital inputs.

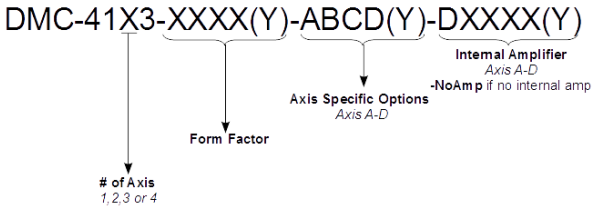
Ordering Options

The DMC-41x3 controller board comes in two sizes, 1-4 axis models (labeled A-D) and 5-8 axis models (labeled E-H). The number of axis is designed by x in the part number DMC-41x3. In addition, Axis A-D and Axis E-H have their own set of axis-specific options that can be ordered. For example, Axis A-D can have a different set of feedback options as Axis E-H even though they reside on the same DMC-41x3 board. The DMC-41x3 can also be ordered with optional internal amplifiers, labeled as AMP or SDM. These amplifiers are "sandwiched" to the back of the DMC-41x3 board. The abstract internal layout of a DMC-41x3 with optional AMP/SDM is shown for 1-4 axis on in the figure below.

1-4 axis DMC-41x3 model (left); 5-8 axis DMC-41x3 model (right)

The full DMC-41x3 part number is a combination of the DMC controller part number (DMC-41x3), for factor ("-XXXX(Y)), axis-specific options (-ABCD(Y) and EFGH(Y)), and optional amplifier types (-DXXXX(Y)), where Y is customization options for that specific board or set of axis. The layout of the full DMC-41x3 part number is shown in the figure below.



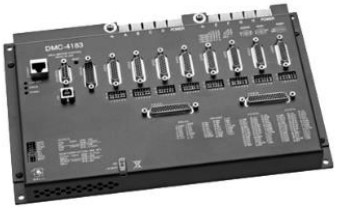


Use the Part Number Generator for building your DMC-41x3
http://www.galil.com/order/part-number-generator/dmc-41x3

Example Part Numbers	
DMC-4143-BOX4(DIN)-ABCD(LSNK)-D3020(HALLF)	4-axis Ethernet/USB Metal Enclosure for 1-4 axis models with DIN Two 500W servo drives • HALLF (provides filter on HALL inputs) 1-8 digital outputs: 25mA, 5-24V opto-isolated, sinking
DMC-4173-BOX8(ETL)-ABCD(HSRC)-EFGH(HSRC)-D4020-D3240	7-axis Ethernet/USB Metal Enclosure for 5-8 axis models Two 1.4 A stepper motor drives-Full, Half, 1/4, 1/16 Four 750W servo drives • ETL 1-8 digital outputs: 500mA 12-24V, sourcing 9-16 digital outputs: 500mA 12-24V, sourcing

DMC-41x3

DMC-41x3 – XXXX(Y) – ABCD(Y) – DXXXX(Y)



Default 1-4 axes model (left), default 5-8 axes model (right)

DMC-41x3 Options

Part Number	Description
DIN	DIN Rail Mount
12V	Power Controller with 12 V_{DC}
16bit	16-bit analog inputs
4-20mA	4-20mA analog inputs
ISCNTL	Isolate Controller Power
TRES4	Encoder terminating resistors, axis 1-4
TRES8	Encoder terminating resistors, axis 5-8
ETL	ETL certification
MO	Motor off jumper installed by default
PCS	Power Cable Set
422	RS-422 on Aux (P2)

	AMP-430x0 (-D30x0)	AMP-43140 (-D3140)	AMP-43240 (-D3240)	AMP-435x0 (-D35x0) AMP-435x7 (-D35x7)	AMP-43640 (-D3640)
Motor type	Brushed/ 3ϕ Brushless servo	Brushed Servo	Brushed/ 3ϕ Brushless servo	Brushed/ 3ϕ Brushless servo	3ϕ Brushless servo
Amplifier Axes	4 or 2	4	4	4 or 2	4
Current Drive	PWM	Linear	PWM	PWM	Linear
Drive Mode	Chopper, Inverter	Linear	Chopper	Phase Shift	Linear
Commutation	Trap w/120° halls	Brushed only	Trap w/120° halls	Sinusoidal	Sinusoidal
Power per axis (Watts per channel)	500	20	750	600	20
Cont. Current (Amps)	7	1	10	8	1
Peak Current (Amps)	10	1	20	15	2
Bus Voltage (VDC)	20 or 80[1]	+/- 12-30 bipolar	20-80[1]	20-80[1]	15-40
Gains (A/V)	0.4, 0.7, 1.0	0.01[2], 0.1	0.5, 1.0, 2.0	0.4, 0.8, 1.6	0.2
Switching Freq. (kHz)	60 or 140[3]	–	24	33	–
Max Current loop BW (kHz)[4]	8	10	3	4	8
Min. Inductance (mH)	0.2 - 0.5	.05	0.8	0.5	.05
Over-Voltage	Yes	No	Yes	Yes	No
Under-Voltage	Yes	No	Yes	Yes	No
Over-Current	Yes	Fused	Yes	Yes	Fused
Short Circuit	Yes	Fused	Yes	Yes	Fused
Over-Temperature	Yes	Thermal Shutdown	Yes	Yes	Thermal Shutdown
ELO	Yes	Yes	Yes	Yes	Yes
Adjustable Current Loop	Yes	No	Yes	Yes	No
Shunt Option	Yes	No	Yes	Yes	Yes
SSR Option	No	Yes	No	No	No

[1] Contact Galil for more options.
[2] Available by ordering the 100mA option.
[3] Contact Galil regarding the 140kHz option.
[4] Current loop bandwidth is system dependent. These values are what can be typically expected.
[5] 0.75 mH @ 24 V_{DC} bus voltage and 1.2 mH minimum @ 48 V_{DC} bus voltage
[6] 0.2 mH when using chopper mode, 0.5 when using inverter mode

SDM Modules

The following embedded stepper amplifier drives are in the same black box as the DMC. Like our servo options, they are available in banks of 2 or 4-axes; note the 2-axes options take up the same space as a bank of 4-axes.

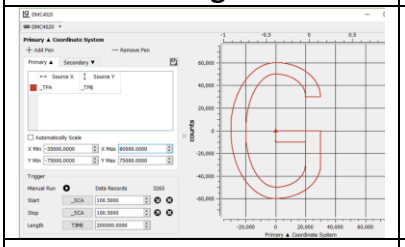
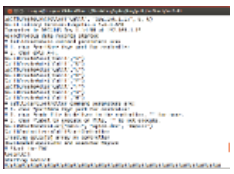
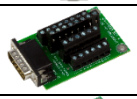
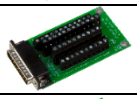
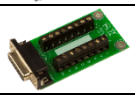
	SDM-440x0 (-D40x0)	**SDM-44140 (-D4140)**	**AMP-D35x7 (-X35x7)**
Motor type	Stepper	Stepper	Stepper
Amplifier Axis	Bank of 2 or 4 axis	Bank of 4 axis	Bank of 2 or 4 axis
Microstepping	$1, \frac{1}{2}, \frac{1}{4}, \frac{1}{16}$	$\frac{1}{64}$	$\frac{1}{256}$
Power per axis	42 W	180 W	480 W
Peak Current	1.4 A/ϕ	3.0 A/ϕ	6.0 A/ϕ
Bus Voltage	12-30 V_{DC}	20-60 V_{DC}	20-80 V_{DC}
Gains	0.5, 0.75, 1.0, 1.4	0.5, 1.0, 2.0, 3.0	0.75, 1.5, 3.0, 6.0
Switching Freq.	27 kHz (nominal)	60 kHz	33 kHz
Min. Inductance	0.5 mH	0.5 mH	0.5 mH
Over-Voltage	No	No	Yes
Under-Voltage	No	Yes	Yes
Over-Current	Yes	Yes	Yes
Short Circuit	Yes	Yes	Yes
Over-Temperature	No	No	Yes
ELO	Yes	Yes	Yes
Low Current Mode (LC)	Yes	Yes	Yes

AMP/SDM Options

The following options can apply to both our servo and stepper (AMP/SDM) modules.

Part Number	**Description**
100mA	100mA output capacity for AMP-43140 Default is 1 Amp
HALLF[1]	Filtered hall sensors
SSR[1]	Solid state relay
ISAMP	Isolates power between amplifiers (two banks of AMP/SDMs required)

1 Not available for all amplifier options

Accessories		
Image	**Part Number**	**Description**
	GDK SOFTWARE	Servo Tuning and Analysis with Terminal, Program Editor, Debugger and Setup tools
	EPICS SOFTWARE	Communication Drivers and Device Support to create a Galil EPICS IOC
	PSR-12-24	12A-24 VDC Power supply
	PSR-6-48	6A-48 VDC Power Supply
	BLM-N23-50-1000-B	Nema 23 Brushless Motor with 1000-line encoder
	CABLE-26-1M	26-pin HD male D to discrete wires, 1 meter
	CABLE-44M-1M	44-pin HD male D to discrete wires-1 meter
	CABLE-44M-2M	44-pin HD male D to discrete wires-2 meter
	CABLE-USB-2M	2 meter USB 2.0 A male to B male cable
	CABLE-USB-3M	3 meter USB 2.0 A male to B male cable
	ICS-48026-M	26-pin D HD male to screw terminals
	ICS-48044-M	44-pin D HD male to screw terminals
	ICS-48115-F	15-pin D LD female to screw term